
ME 637: Wind Power Plant

Course Code	ME 637
Course Name	Wind Power Plant
L-T-P-C	3-0-0-3
Prerequisites	Instructor Consent
Intended for	UG/M. Tech. /MS/PhD
Distribution	Specialized stream elective for M. Tech in Mechanical Engineering with specialization in Energy Engineering, elective for other students
Approval	10th Senate (Jun 2016)

Course Contents

- **Module I**

Introduction, General theories of wind machines, Basic laws and concepts of aerodynamics [8 Lectures]

- **Module II**

Micro-siting, Description and performance of the horizontal-axis wind machines, Blade design, Description and performance of the vertical-axis wind machines [12 Lectures]

- **Module III**

The generation of electricity by wind machines, case studies, overview of micro, mini and small hydro, site selection and civil works, penstocks and turbines, speed and voltage regulation. [14 Lectures]

- **Module IV**

Investment issues, load management and tariff collection, distribution and marketing issues, case studies. [8 Lectures]

Text Books

1. Gasch Jochen Twele, **Wind Power Plants Fundamentals, Design, Construction and Operation**, Springer-Verlag Berlin, 2012
2. Mathew Sathyajit, **Wind Power Plants: Fundamentals, Resource Analysis and Economics**, Springer-Verlag Berlin, 2006.